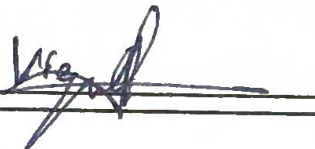


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)		
Student Name:	Vignesh P. S	Reg. No.:	192511177
Section / Batch:		Date:	

Code of Conduct

I, Vignesh P. S (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
GRAND TOTAL:			50 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark	Mark
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

8

6

1

- a)
- ① To Scan, Analyze, and understand Images
 - ② To Recognize Objects and Support Decision Making.

- b)
- ① Face recognition
 - ② Self driving (Autonomous) Vehicles

1)

a) Feature extraction is the process of Identifying Important Information Such as edges, corners, etc.

- b)
- ① It helps Identify and classify objects Accurately Making System more Intelligent and Reliable

- 3)
- a) It Involves basic operation Like noise Removal Image enhancement and filtering
- b) Mid level : Image Segmentation and feature extraction.

b) Illumination provide light to capture the Image.

8.

a) Image processing Improves Image quality

b) Image processing edits Images Computer Vision understands Images

9)

a) In healthcare, It's help detect diseases from x-rays

b) In Self-driving cars, It detects roads and obstacles.

10) a) pixels are the Basic parts of an Image

b) Higher resolution and good Sampling given Better Image Analysis.

3)

a)

b)